



(KTXFI2EBNF) Physics II. Supplementary Material

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Justifying of this Material I

- ▶ We did not have sufficient time to cover the following topics:
 - ▶ Basic concepts of nuclear physics.
 - ▶ Basic concepts of particle physics.
- ▶ I can only briefly outline what these topics encompass: radioactivity, isotopes, neutron radiation, radiation protection, neutron-induced nuclear fission, thermonuclear fusion, nuclear batteries, types of nuclear reactors, thermonuclear experiments, nuclear weapons, thermonuclear weapons, nuclear waste management, the Hungarian Nuclear Power Plant in Paks, and future technologies such as advanced reactors, high-temperature reactors, thorium reactors, and modular reactors.
- ▶ What could I meaningfully tell you about particle physics at this point? For example: hadrons, mesons, leptons, quarks, or particle colliders?

Justifying of this Material II

- ▶ We have run out of time; therefore, I will send you additional links, or I may record and upload a short personal video exclusively for you.
- ▶ Naturally, none of these topics will be included in the midterm or final examinations.

Basic concepts of nuclear physics I

- ▶ Radioactivity, isotopes, neutron radiation, radiation protection:
 - ▶ Radioactivity
 - ▶ Nuclear Power Plant in Paks
 - ▶ Neutron Radiation
 - ▶ Radiation Protection
- ▶ Neutron-induced nuclear fission, thermonuclear fusion, nuclear batteries, types of nuclear reactors, thermonuclear experiments, nuclear weapons, thermonuclear weapons, nuclear waste management:
 - ▶ Chain Reaction
 - ▶ Nuclear Fusion Experiments
 - ▶ Stellar Nuclear Fusion
 - ▶ Natural Fission Reactor
 - ▶ Nuclear Weapons
 - ▶ Radioisotope Thermoelectric Generator (RTG)

Basic concepts of nuclear physics II

- ▶ Radioactive Waste Management
- ▶ Oppenheimer
- ▶ Dr. Strangelove or: How I Learned to Stop Worrying and Love the Bomb
- ▶ Future technologies such as advanced reactors, high-temperature reactors, thorium reactors, and modular reactors:
 - ▶ Advanced Nuclear Reactors
 - ▶ Thorium Reactors
 - ▶ Small Modular Reactors

Basic concepts of particle physics

- ▶ Hadrons, Mesons, Leptons, Quarks
- ▶ Conseil Européen pour la Recherche Nucléaire (CERN)

The End

Thank you for your attention!